

ϕ TD-Gaussian

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.04809	0.04739	0.05248	0.0773
Cramér $\ell_2^2 \downarrow$	0.0003212	0.0002763	0.0005268	0.0007295
CF - $\ell_2^2/\omega^2 \downarrow$	0.0002227	0.0002168	0.000478	0.0004967

ϕ TD-Cauchy

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1164	0.03845	0.0734	0.1308
Cramér $\ell_2^2 \downarrow$	0.003667	0.0001854	0.001672	0.00241
CF - $\ell_2^2/\omega^2 \downarrow$	0.002364	0.0001583	0.0008483	0.001229

ϕ TD-Gamma

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.7161	2.494	1.951	0.07296
Cramér $\ell_2^2 \downarrow$	0.2972	0.7989	0.7373	0.0003444
CF - $\ell_2^2/\omega^2 \downarrow$	0.2614	0.7255	0.6958	0.0002273

ϕ TD-Categorical

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2816	0.2857	0.2931	0.2862
Cramér $\ell_2^2 \downarrow$	0.028	0.01464	0.03496	0.02896
CF - $\ell_2^2/\omega^2 \downarrow$	0.0207	0.01159	0.02869	0.02291

ϕ TD-Quantile

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.08326	0.1839	0.1081	0.1247
Cramér $\ell_2^2 \downarrow$	0.00118	0.001824	0.002871	0.001758
CF - $\ell_2^2/\omega^2 \downarrow$	0.0007196	0.002734	0.002336	0.001283